

Bivariate Data

Introduction

Sometimes, in practical applications, we might come across certain set of data, where each item of the set may comprise of the values of two or more variables.

A bivariate data is a set of paired measurements which are of the form $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

Examples include:

- i. Marks obtained in two subjects by 60 students in a class.
- ii. The series of sales revenue and advertising expenditure of the various branches of a company in a particular year.
- iii. The series of ages of husbands and wives in a sample of selected married couples.

Our interest is to find a relationship (if it exists) between the two variables under study.

Scatter Diagrams and Correlation

A scatter diagram is a tool for analysing relationships between two variables.

One variable is plotted on the horizontal axis and the other is plotted on the vertical axis.

The pattern of their intersecting points can graphically show relationship patterns.

Most often a scatter diagram is used to prove or disprove cause-and-effect relationships. While the diagram shows relationships, it does not by itself prove that one variable causes the other.

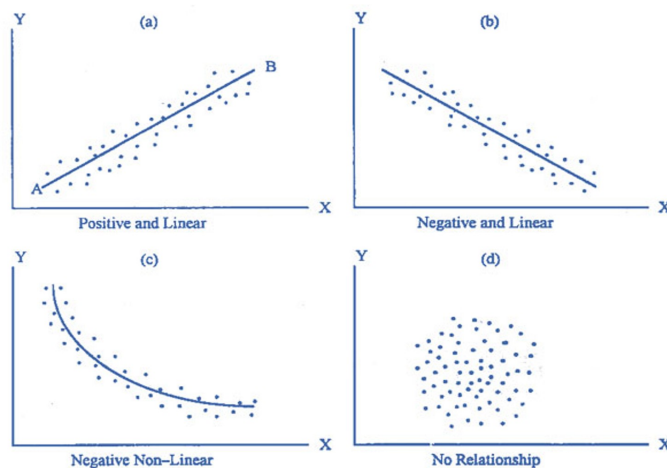
In brief, the easiest way to visualize Bivariate Data is through a Scatter Plot. "Two variables are said to be correlated if the change in one of the variables results in a change in the other variable".

Positive and Negative Correlation

If the values of the two variables deviate in the same direction i.e. if an increase (or decrease) in the values of one variable results, on an average, in a corresponding increase (or decrease) in the values of the other variable the correlation is said to be positive.

Correlation between two variables is said to be negative or inverse if the variables deviate in opposite direction.

That is, if increase (or decrease) in the values of one variable results in corresponding decrease (or increase) in the values of the other variable.



Interpreting a Scatter Plot

Scatter diagrams will generally show one of six possible correlations between the variables:

- Strong Negative Correlation** The value of Y clearly decreases as the value of X increases.
- Weak Positive Correlation** The value of Y increases slightly as the value of X increases.
- Weak Negative Correlation** The value of Y decreases slightly as the value of X increases.
- Complex Correlation** The value of Y seems to be related to the value of X, but the relationship is not easily determined.
- Strong Positive Correlation** The value of Y clearly increases as the value of X increases.
- No correlation** There is no demonstrated connection between the two variables

Correlation Coefficient

It's a measure of the strength of linear association between 2 paired variables It takes values from - 1 to +1.

Pearson's Product Moment Correlation Coefficient

Pearson's product moment correlation coefficient, usually denoted by r , is one example of a correlation coefficient.

It is a measure of the linear association between two variables that have been measured on interval or ratio scales, such as the relationship between height and weight.

However, it can be misleadingly small when there is a relationship between the variables but it is a non-linear one.

The correlation coefficient r is given by:

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}}$$

It can also be shown that:

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

Example 1

Obtain the correlation coefficient of the following data

Mean Temp (x)	14.2	14.3	14.6	14.9	15.2	15.6	15.9
Pirates (y)	35000	45000	20000	15000	5000	400	17

Solution

Mean Temp. (x)	Pirates (y)	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
14.2	35000	-0.76	17797.57	0.57	316753548	-13475
14.3	45000	-0.66	27797.57	0.43	772704977	-18266
14.6	20000	-0.36	2797.57	0.13	7826405	-999
14.9	15000	-0.06	-2202.43	0	4850691	125
15.2	5000	0.24	-12202.43	0.06	148899263	-2963
15.6	4000	0.64	-16802.43	0.41	282321605	-10801
15.9	17	0.94	-17185.43	0.89	295338955	-16203
Tot. = 104.7	120417	0	0	2.5	1828695447	-62583
$\bar{x} = 14.96$	$\bar{y} = 17202.43$			S_{xx}	S_{yy}	S_{xy}

We then have $r = \frac{-62583}{\sqrt{2.5(1828695447)}} \approx -0.93$

In R the following can be achieved as:

```
# reading in the data

x <- c(14.2, 14.3, 14.6, 14.9, 15.2, 15.6, 15.9)

y <- c(35000, 45000, 20000, 15000, 5000, 4000, 17)

cor(x, y, method = "pearson")
```

[1] -0.9199436

Example 2

A study was conducted to find whether there is any relationship between the weight and blood pressure of an individual. The following set of data was arrived at from a clinical study. Let us determine the coefficient of correlation for this set of data. The first column represents the serial number and the second and third columns represent the weight and blood pressure of each patient.

Weight	78	86	72	82	80	86	84	89	68	71
Blood Pressure	140	160	134	144	180	176	174	178	128	132

Solution

Totals

X	78	86	72	82	80	86	84	89	68	71	796
Y	140	160	134	144	180	176	174	178	128	132	1546
xy	10920	13760	9648	11808	14400	15136	14616	15842	8704	9372	124206
x^2	6084	7396	5184	6724	6400	7396	7056	7921	4624	5041	63826
y^2	19600	25600	17956	20736	32400	30976	30276	31684	16384	17424	243036

$$\text{Thus } r = \frac{10(124206) - (796)(1546)}{\sqrt{[(10)63776 - (796)^2][10(243036) - (1546)^2]}} = \frac{11444}{\sqrt{(1144)(40244)}} = 0.5966$$

Spearman's rank correlation coefficient

Data which are arranged in ascending order are said to be in ranks or ranked data.

The coefficient of correlation for such type of data is given by Spearman rank difference correlation coefficient and is denoted by R.

R is given by the formula:

$$R = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

Example

The data given below are obtained from student records. (Grade Point Average (x) and Graduate Record exam score (y)) Calculate the Spearman's rank correlation coefficient.

Solution

Subject	1	2	3	4	5	6	7	8	9	10
X	8.3	8.6	9.2	9.8	8.0	7.8	9.4	9.0	7.2	8.6
y	2300	2250	2380	2400	2000	2100	2360	2350	2000	2260

Solution \ Note that in the x row, we have two students having a grade point average of 8.6 also in the y row; there is a tie for 2000.

Now we arrange the data in descending order and then rank 1, 2, 3, ..., 10 accordingly. In case of a tie, the rank of each tied value is the mean of all positions they occupy. In x , for instance, 8.6 occupy ranks 5 and 6. So each has a rank 5.5.

Similarly in ' y ' 2000 occupies ranks 9 and 10, so each has rank 9.5.

Now we come back to our formula. We compute d , square it and substitute its value in the formula:

X	8.3	8.6	9.2	9.8	8.0	7.8	9.4	9.0	7.2	8.6
y	2300	2250	2380	2400	2000	2100	2360	2350	2000	2260
Rank(X)	7	5.5	3	1	8	9	2	4	10	5.5
Rank(Y)	5	7	2	1	9.5	8	3	4	9.5	6
d	2	-1.5	1	0	-1.5	1	-1	0	0.5	-0.5
d²	4	2.25	1	0	2.25	1	1	0	0.25	0.25

So here, $n = 10$ and $\sum d^2 = 12$. So:

$$R = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} = 1 - \frac{6(12)}{10(10^2 - 1)} = 1 - \frac{72}{990} \approx 0.9273$$

Note: If we are provided with only ranks without giving the values of x and y we can still find Spearman rank difference correlation R by taking the difference of the ranks and proceeding in the above shown manner.

In R the above can be obtained as:

```
# 1. Define the data vectors
x <- c(8.3, 8.6, 9.2, 9.8, 8.0, 7.8, 9.4, 9.0, 7.2, 8.6)
y <- c(2300, 2250, 2380, 2400, 2000, 2100, 2360, 2350, 2000, 2260)

cor(x, y, method = "spearman")
```

```
[1] 0.9268293
```

Regression

If two variables are significantly correlated, and if there is some theoretical basis for doing so, it is possible to predict values of one variable from the other.

Regression analysis, in general sense, means the estimation or prediction of the unknown value of one variable from the known value of the other variable.

Regression Equation

Regression analysis can be thought of as being sort of like the flip side of correlation.

It has to do with finding the equation for the kind of straight lines you were just looking at.

Suppose we have a sample of size n and it has two sets of measures, denoted by x and y .

The simple regression with only one independent variable is given as:

Simple Linear Regression

The **simple linear regression** is a model with a single regressor x that has a relationship with a response y that is a straight line.

The simple linear regression model is given by:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

where β_0 is the intercept and β_1 is the slope and are unknown constants and ϵ is a random error component.

The errors:

- Assumed to have mean zero and unknown variance σ^2
- The errors are uncorrelated meaning that the value of one error does not depend on the value of the other error.

The response Y is a random variable with the mean of the distribution given as:

$$E(Y|X) = \beta_0 + \beta_1 X$$

and the variance given as

$$Var(Y|X) = Var(\beta_0 + \beta_1 X + \epsilon) = \sigma^2$$

The parameters β_0 and β_1 are usually called the **regression coefficients**.

The slope β_1 is the change in the mean distribution of Y produced by a unit change in X

The intercept β_0 is the mean distribution of the response Y when $X = 0$

Least Squares Estimation of the Parameters

Suppose that we have n pairs of data say $(y_1, x_1), (y_2, x_2), \dots, (y_n, x_n)$

Estimation of β_0 and β_1

The **method of least squares** is used to estimate β_0 and β_1 .

We may write the regression model as:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, i = 1, 2, \dots, n$$

We minimize the squared errors given by:

$$\sum_{i=1}^n \epsilon_i^2 = S(\beta_0, \beta_1) = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2$$

We differentiate the function partially wrt to β_0 and β_1 and equate to zero and we have

$$\frac{\partial S}{\partial \beta_0} = -2 \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$$

$$\frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = 0$$

Simplifying these two equations we have:

$$n\hat{\beta}_0 = \hat{\beta}_1 \sum_{i=1}^n X_i = \sum_{i=1}^n Y_i$$

$$\hat{\beta}_0 \sum_{i=1}^n X_i + \hat{\beta}_1 \sum_{i=1}^n X_i^2 = \sum_{i=1}^n Y_i X_i$$

The above two equations are called **least-squares normal equations** and the solutions to the equations are given as:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \quad (1)$$

and

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n Y_i X_i - n\bar{x}\bar{y}}{\sum_{i=1}^n X_i^2 - n\bar{x}^2} \quad (2)$$

$\hat{\beta}_0$ and $\hat{\beta}_1$ in Equations 1 and 2 are the **least-squares estimators** of the intercept and slope.

The fitted simple linear regression model is then

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X \quad (3)$$

The difference between the observed value Y_i and the corresponding fitted value $\hat{Y}_i$ is the **residual** defined as

$$e_i = Y_i - \hat{Y}_i, i = 1, 2, \dots, n \quad (4)$$

Example 1 Consider the data below and fit a simple linear regression model

Table 1: Example 1 data

Y_i	2158.70	1678.15	2316	2061.30	2207.50	1708.30	1784.70	1784.70	2575.00	2357.90
X_i	15.50	23.75	8.00	17.00	5.50	19.00	24.00	2.50	7.50	11.00

Solution

To estimate the model calculate

$$\sum_{i=1}^n Y_i X_i = 264028.9, \sum_{i=1}^n X_i^2 = 2308.062, \bar{X} = 13.375, \bar{Y} = 2063.225$$

Making the appropriate substitution we have

$$\hat{\beta}_1 = \frac{264028.9 - 10 \times 13.375 \times 2063.225}{2308.062 - 10 \times 13.375^2} = -22.97469$$

$$\hat{\beta}_0 = 2063.225 - (-22.97469 \times 13.375) = 2370.51$$

Thus the fitted line is given as:

$$\hat{Y} = 2370.51 - 22.975X \quad (5)$$

One can obtain the fitted values and the residuals of the model using previously defined equations.

Example using R We can fit the model using R and obtain the same results we got by hand.

We start by reading in the data as follows:

```
# reading in the data
Y <- c(2158.70, 1678.15, 2316, 2061.30, 2207.50, 1708.30, 1784.70, 1784.70, 2575, 2357.90)

X <- c(15.50, 23.75, 8.00, 17, 5.5, 19, 24, 2.5, 7.5, 11)
```

We fit the simple linear model and obtain the fitted model results in form of a table

```
# reading in the data
Y <- c(2158.70, 1678.15, 2316, 2061.30, 2207.50, 1708.30, 1784.70, 1784.70, 2575, 2357.90)

X <- c(15.50, 23.75, 8.00, 17, 5.5, 19, 24, 2.5, 7.5, 11)

# fitting the model

model <- lm(Y~X)
knitr::kable(broom::tidy(model))
```

term	estimate	std.error	statistic	p.value
(Intercept)	2370.51090	182.29049	13.004030	0.0000012
X	-22.97465	11.99887	-1.914735	0.0918583